
The Shadow-Software Toll Scam

How cash kept flowing through India's cashless highways

CashlessConsumer Research

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How cash kept flowing through India's "cashless" highways — and who looked away

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In one sentence: At roughly 100 Indian highway toll plazas, operators installed unauthorised “shadow” software on their own booth computers that printed real-looking receipts for cash taken from non-FASTag vehicles — while showing the national toll system almost none of it. The fraud ran for about two years across at least 14 states before a police sting, a parliamentary admission, a botched crackdown, a court reversal and finally a federal money-laundering probe put the spotlight back on it.

Why you should read this

If you have ever driven on an Indian national highway, this case is about a system you paid into — and about the small fraction of payments that still happen outside the digital trail. It explains:

1. **How tolling actually works** — the hardware, the software, and the little-known manual side-door that still existed at every plaza.
2. **How the bypass worked** — in plain language, reconstructed from police and enforcement-directorate findings.
3. **Why the audits missed it** — a systems failure worth understanding, because it will happen again in other digital-public-infrastructure systems.
4. **What happened to accountability** — including the part where the High Court undid the regulator's punishment, and what happened next.
5. **What the open data shows** — we matched the scam plazas against the government's own plaza registry, and the results are uncomfortable.
6. **What changes now** — and what risks are simply migrating to the next system.

This report is written for a reader with **zero background** in tolling systems or this case. Every technical claim is sourced; every unverified claim is labelled as such. A full evidence archive underlies this report (see Appendix C).

Part I — The system, before the scam

1. Toll plazas 101: what actually happens at the booth

India's national highways (the "NH" network) are largely funded through **user fee** — the toll you pay at a plaza. There were, as of the government's own registry from August 2025, **more than 1,100 fee plazas** across 24 states and union territories (we count them in Part IV).

A toll plaza is not run by the government. It is run by a **fee-collection agency** — a private contractor engaged by the concessionaire (the company that built or operates the highway section) under the overall supervision of the **National Highways Authority of India (NHAI)**, a statutory body under the Ministry of Road Transport and Highways (MoRTH). A separate NHAI joint venture, the **Indian Highways Management Company Limited (IHMCL)**, runs the technology backbone of electronic tolling.

At the booth, the stack looks like this:

Layer	What it is	Who controls it
Lane hardware	RFID (FASTag) reader, barrier arm, vehicle-classification sensors, CCTV/ANPR camera, thermal receipt printer	The concessionaire's vendor
Toll Management System (TMS)	The lane software: reads the tag, classifies the vehicle, computes the fee, records the transaction, prints the receipt	The fee-collection agency's technology vendor
Back office	Batch settlement of the day's transactions to the acquiring bank, then to IHMCL/NPCI's NETC clearing system	Agency + bank
Oversight	NHAI Regional Offices inspect plazas; IHMCL monitors NETC transactions; ministry committees review	Government

The critical sentence in that table is the third row of the first column: **the TMS lane software is procured and controlled by the private collection agency, not by NHAI**. Hold that thought.

2. FASTag and the two kinds of traffic

Since February 2021, paying toll on national highways by cash has been formally the exception, not the rule. Vehicles carry a **FASTag** — an RFID sticker linked to a prepaid wallet or bank account. As you approach the plaza:

- The reader identifies the tag, the lane software computes the fee for your vehicle class, the barrier lifts, your account is debited, and the transaction flows into the **National Electronic Toll Collection (NETC)** ecosystem operated by NPCI (the same organisation behind UPI).
- Radio-frequency tags fail sometimes — dead battery in the reader, damaged sticker, insufficient balance, or a tag listed as blacklisted. For exactly these vehicles, the barrier stays down and the booth operator keys the vehicle class manually, charges a **double fee** (the “2× penalty” for travelling without a valid tag), prints a receipt and accepts **cash**.

In ministerial statements to Parliament, roughly **98% of transactions** were electronic; the non-FASTag residue was about **2% of vehicle volume**¹. Two per cent sounds trivial. It is not. Across a national network handling hundreds of millions of trips, the non-FASTag lane is a constant, cash-heavy, operator-controlled stream — and, as this case shows, it was being **skimmed at industrial scale**.

3. The manual side-door

Here is the design weakness at the heart of this whole story, and it was visible in official documents years before the scam surfaced:

- The operator’s keystroke in the TMS decides what the national system believes happened at the lane. If the operator marks a paying vehicle as **“exempted”** (exempt category) or **“violation”** (jumped the barrier, or could not be charged), no toll is recorded as due.
- There was **no independent physical count** at the lane — no reconciling of barrier events or sensor counts against keyed transactions.
- The agency’s own technology vendor controlled the lane software that produced the receipt and the record.

¹<https://www.gstseat.com/news/parliament-standing-committee-on-transparency-in-toll-collection-detailed-answer-given-by-minister-gadkari-in-lok-sabha/>

- A program to monitor whether agencies actually **remit** what they collect (“Toll Remittance Alert System”) was only being tendered in April 2023 — *after* the fraud had already begun².

In short: the national system trusted the lane, and the lane belonged to the contractor.

²IHMCL, “Selection of Agency for Design, Development, Implementation... of Toll Remittance Alert System (TRAS)”, RFP, 3 April 2023 (PDF in evidence archive).

Part II — The scam

4. January 2025: a sting at one plaza blows open a network

On 22 January 2025, Uttar Pradesh police's Special Task Force (STF) registered FIR No. 0017/2025 at Lalganj police station, Mirzapur district, and moved on the **Atraila Shiv Gulam toll plaza** on the Varanasi–Hanumana section (old NH-7, now NH-135) — a plaza on the pilgrimage-and-coal corridor southeast of Mirzapur³.

Within two days the STF had:

- Arrested **Alok Kumar Singh**, 35, an MCA (computer-applications) graduate from Jaunpur district who had briefly worked at a toll plaza in 2021. He was picked up near Babatpur airport, Varanasi⁴.
- Detained the plaza's manager, **Rajeev "Raju" Mishra** of Prayagraj, and an operator, **Manish Mishra** of Sidhi (MP)⁵.
- Seized two laptops, a printer, five mobile phones, ₹19,580 in cash and a vehicle⁶.

What the STF described was not a cash-skimming trick by a rogue operator. It was a **software product**, allegedly built by Singh, that had been **installed on the plaza's own booth computers** and operated in parallel with the official TMS⁷.

5. How the bypass worked — the reconstruction

Pulling together STF briefings, accused interrogations and later ED disclosures, the machine had these parts:

- 1. The trigger.** A vehicle arrives without a valid FASTag. Barrier stays down. Under the rules, the operator must charge double the fee, in cash, and the transaction must be keyed into the official TMS.
- 2. The shadow app.** On the booth computer, alongside the TMS, ran an unauthorised application (referred to in ED statements by the names '**Mobdata**' and '**Any**' — see the caveat in §11). It computed

³<https://timesofindia.indiatimes.com/city/lucknow/mca-grad-behind-toll-fraud-arrested/articleshow/117498101.cms>

⁴<https://timesofindia.indiatimes.com/city/lucknow/mca-grad-behind-toll-fraud-arrested/articleshow/117498101.cms>

⁵<https://www.aajtak.in/uttar-pradesh/story/toll-plaza-software-scam-fraud-worth-crores-stf-caught-3-accused-varanasi-uttar-pradesh-lclar-strc-2149511-2025-01-23>

⁶<https://www.opindia.com/2025/01/nhai-orders-action-in-the-toll-plaza-scam-uncovered-by-the-up-stf-3-arrested/>

⁷<https://www.bhaskar.com/local/up/lucknow/news/toll-plaza-scam-3-gang-members-detained-including-software-engineer-133802942.html>

the **same double fee** from the public gazette fee schedules, so the receipt it printed was arithmetically indistinguishable from a genuine one⁸.

3. The forged receipt. The app used the **same thermal printer path** as the TMS. The driver got a plausible receipt showing the toll and the 2× penalty. Nothing the driver could see was wrong⁹.

4. The vanish. Only about **5%** of these non-FASTag transactions were keyed into the official TMS — enough to keep the “exempted/violation” categories looking ordinary. The other 95% existed only inside the shadow app¹⁰.

5. The monitoring layer. Plaza-level collections were pushed to a **central monitoring system** — ED describes separate monitoring portals; STF found Singh’s laptop had direct access to plaza data across plazas¹¹¹².

6. The split. Cash stayed at the plaza level; Singh’s share moved through at least five bank accounts and digital wallets, according to ED findings reported in September 2026¹³.

Why this was invisible from above

Do the arithmetic the way an auditor would. Non-FASTag traffic is about 2% of volume. Skimming 95% of it removes **under 2% of gross revenue** — a rounding error in aggregate charts, easily masked by genuine monthly variance. And every audit trail NHAI could query — the TMS database, the NETC settlement records, the bank remittances — was fed by the very lane software the shadow app sat beside. **The fraud never wrote to the system that was being audited.**

That is why, when NHAI’s technical team examined computers at plazas in Gorakhpur after the STF raids, it reportedly found “no significant evidence”¹⁴. Auditing the books cannot catch a second set of books.

6. The footprint: from one plaza to a hundred

⁸<https://navbharattimes.indiatimes.com/metro/lucknow/crime/toll-tax-fraud-by-fake-receipts-through-software-at-42-city-of-india-news/articleshow/117503938.cms>

⁹<https://navbharattimes.indiatimes.com/metro/lucknow/crime/toll-tax-fraud-by-fake-receipts-through-software-at-42-city-of-india-news/articleshow/117503938.cms>

¹⁰<https://www.opindia.com/2025/01/nhai-orders-action-in-the-toll-plaza-scam-uncovered-by-the-up-stf-3-arrested/>

¹¹<https://www.opindia.com/2025/01/nhai-orders-action-in-the-toll-plaza-scam-uncovered-by-the-up-stf-3-arrested/>

¹²<https://english.mathrubhumi.com/news/crime/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-without-fastag-ed-o0qf4hpy>

¹³<https://timesofindia.indiatimes.com/city/lucknow/ed-conducts-searches-at-14-premises-in-6-states-seizes-rs-1-03-crore-cash-in-toll-collection-fraud-case/articleshow/133732293.cms>

¹⁴Hindustan Times, Gorakhpur edition, January 2025 (press capture in evidence archive).

Date	Who says	Count
23 Jan 2025	Day-1 interrogation claims	“up to 200 plazas” (confession-based, never confirmed) ¹⁵
23 Jan 2025	STF verified list	42 plazas in 14 states , operators named ¹⁶
13 Feb 2025	STF	150–200 plazas “under examination”; notices to managers in 6 districts ¹⁷
3 Sep 2026	ED (device forensics)	~100 plazas where the apps were used ¹⁸

The verified pattern: 42 was the *arrest perimeter*, not the ceiling. ED’s forensic examination of seized devices in 2026 nearly tripled the confirmed footprint, and pointed to plazas in **UP, Rajasthan, Madhya Pradesh, Jammu & Kashmir, West Bengal and the Delhi-NCR** among the searched premises¹⁹²⁰.

7. The money

- **Per plaza:** STF’s working estimate at Atraila was roughly **₹45,000 per day** of skimmed cash²¹. Over the ~730 days the fraud reportedly ran, that is about **₹3.3 crore from a single plaza** — a figure Dainik Bhaskar separately reported as the “3.28 crore embezzled at one location”²².

¹⁵<https://www.bhaskar.com/local/up/lucknow/news/toll-plaza-scam-3-gang-members-detained-including-software-engineer-133802942.html>

¹⁶<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

¹⁷<https://www.amarujala.com/uttar-pradesh/mirzapur/notice-to-toll-plaza-managers-of-six-districts-in-120-crore-rupees-scam-in-mirzapur-2025-02-13>

¹⁸<https://www.tribuneindia.com/news/india/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-not-bearing-fastag-stickers-ed/>

¹⁹<https://timesofindia.indiatimes.com/city/lucknow/ed-conducts-searches-at-14-premises-in-6-states-seizes-rs-1-03-crore-cash-in-toll-collection-fraud-case/articleshow/133732293.cms>

²⁰<https://www.dailyexcelsior.com/about-100-highway-plazas-including-some-in-jk-used-illegal-apps/>

²¹<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

²²<https://www.bhaskar.com/local/up/lucknow/news/toll-plaza-scam-3-gang-members-detained-including-software-engineer-133802942.html>

- **Network scale:** the STF put total diversion at **₹120 crore+** across the 42 plazas²³. Early day-1 media claims reached ₹750 crore — never substantiated, treat with caution.
- **ED’s 2026 language:** receipts for unauthorised collections “running into crores” across ~100 plazas; ₹1.03 crore cash seized in the September 2026 searches and **eight bank accounts** frozen²⁴.

8. The humans

Actor	Role	Status
Alok Kumar Singh	Alleged developer of the shadow software; MCA, ex-toll-plaza worker	Arrested Jan 2025
Rajeev “Raju” Mishra	Manager, Shivgulam plaza (Prayagraj native)	Arrested Jan 2025
Manish Mishra	Booth operator (Sidhi, MP)	Arrested Jan 2025
Saaban Lal Kumhvat	Fourth arrest	Feb 2025 ²⁵
“Sawant” and “Sukhantu”	Alleged system managers for the 42-plaza operation	Named in STF briefing; status unclear ²⁶
“Veltek” (per Dainik Jagran)	Collection agency named for Atraila plaza	Identity unresolved; not among the debarred 14 ²⁷

Publishing caution: several names above come from single-outlet or briefing-based reporting. We label them as alleged, and we do not name private individuals beyond what multiple independent reports establish.

²³<https://www.thedailyjagran.com/india/toll-collection-fraud-nhai-bans-14-agencies-for-using-fake-software-to-collect-toll-illegally-rs-100-cr-security-confiscated-10225098>

²⁴<https://timesofindia.indiatimes.com/city/lucknow/ed-conducts-searches-at-14-premises-in-6-states-seizes-rs-1-03-crore-cash-in-toll-collection-fraud-case/articleshow/133732293.cms>

²⁵<https://www.amarujala.com/uttar-pradesh/mirzapur/notice-to-toll-plaza-managers-of-six-districts-in-120-crore-rupees-scam-in-mirzapur-2025-02-13>

²⁶<https://www.opindia.com/2025/01/nhai-orders-action-in-the-toll-plaza-scam-uncovered-by-the-up-stf-3-arrested/>

²⁷<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

Part III — Accountability, in three acts

9. Act one: the crackdown (March 2025)

The STF case triggered the fastest nationwide contractual crackdown NHAI had executed:

- In Parliament (20–21 March 2025), Minister Nitin Gadkari confirmed the bypass: vehicles without valid tags were being recorded as “exempted or violation” with receipts never entering the system — while insisting there was **no breach of the electronic tolling system itself**²⁸. Announcements followed: AI-based audit cameras and ANPR pilots.
- NHAI **debarred 14 user-fee collection agencies** for one to two years, **encashed performance securities reported at over ₹100 crore**, and levied penalties exceeding ₹2 crore²⁹³⁰.
- Among the debarred: AK Construction (linked to nine plazas on the scam list), RK Jain Infra Projects, Vanshika Construction, Innovision Ltd and others³¹. The Atraila agency was separately terminated with its bank guarantee encashed.

10. Act two: the reversal (July 2026)

In July 2026, the **Delhi High Court (Justice Sachin Datta)** set aside NHAI’s debarments, contract terminations and bank-guarantee encashments³². The court’s reasoning, as reported:

- **No agency employee was named in any charge sheet** — the prosecuted individuals were the software operator’s network, not the contractors’ staff.
- **No shadow software was recovered from the petitioners’ plazas** in any seizure.
- NHAI’s own inspections had found nothing; the action rested on a press note, a confession attributed to a third party, and **statistical inference** (the revenue pattern) — which the court held was not credible material for such penalties.

This is the quiet earthquake of the case. The entire contractual accountability layer — the ₹100 crore in securities, the debarments, the terminations — **collapsed**. Whatever the ultimate truth about what collection agencies knew, the *evidence* assembled through the contractual route did not survive judicial scrutiny.

²⁸<https://www.gstseat.com/news/parliament-standing-committee-on-transparency-in-toll-collection-detailed-answer-given-by-minister-gadkari-in-lok-sabha/>

²⁹<https://money.rediff.com/news/market/nhai-debars-14-toll-agencies-for-irregularities/23868620250321>

³⁰<http://www.uniindia.com/nhai-bans-14-agencies-for-irregular-activities-in-fee-collection-at-toll-plazas/india/news/3418421.html>

³¹<https://money.rediff.com/news/market/nhai-debars-14-toll-agencies-for-irregularities/23868620250321>

³²<https://indialegallive.com/constitutional-law-news/courts-news/delhi-high-court-sets-aside-nhai-action-against-toll-contractors-in-software-fraud-case/>

11. Act three: the money-laundering probe (September 2026)

Which brings us to the news that started this investigation: on **2 September 2026**, the **Enforcement Directorate (ED)** — the federal financial-crime agency — conducted searches at **14 premises across six states** under the Prevention of Money Laundering Act (PMLA), seized ₹1.03 crore in cash and froze eight bank accounts³³. On 3 September, the ED disclosed that its forensic examination of devices pointed to the unauthorised apps and monitoring portals at **about 100 plazas**³⁴³⁵.

Read in sequence, the ED action is the state's **second attempt at accountability** — this time using criminal-forensic machinery (device images, banking trails, portal data) rather than contractual penalties, precisely because the contractual route failed in court.

On the names 'Mobdata' and 'Any': these app names appear only in ED-attributed wire copy. We searched app stores, code-hosting platforms, and certificate-transparency logs; **no public trace of any product by these names exists**. They are likely internal codenames visible on the seized devices and portals. We treat them as attributed claims, not established product names — and we note that “MobData” is also the name of an unrelated Chinese analytics SDK, which should not be conflated.

³³<https://timesofindia.indiatimes.com/city/lucknow/ed-conducts-searches-at-14-premises-in-6-states-seizes-rs-1-03-crore-cash-in-toll-collection-fraud-case/articleshow/133732293.cms>

³⁴<https://www.tribuneindia.com/news/india/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-not-bearing-fastag-stickers-ed/>

³⁵<https://english.mathrubhumi.com/news/crime/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-without-fastag-ed-o0qf4hpy>

Part IV — What the open data shows

This is the part you cannot get from news reports, so we built it.

12. The plaza registry, counted

IHMCL publishes the official list of national-highway fee plazas as a PDF³⁶. We parsed it (methodology in Appendix A) and extracted **1,133 plazas**, each with its NETC plaza code, name, state and district.

Where the plazas are:

State	Plazas
Rajasthan	163
Uttar Pradesh	135
Maharashtra	101
Madhya Pradesh	89
Tamil Nadu	81
Andhra Pradesh	79
Haryana	78
Karnataka	62
Gujarat	56
Punjab	41
Bihar	41
Telangana	36
Odisha	32
West Bengal	29
Chhattisgarh	23
Jharkhand	22
Delhi	13
Assam	10

³⁶IHMCL, “National Highway (NH) Fee Plazas” list, August 2025 (PDF in evidence archive; <https://ihmcl.co.in/>).

State	Plazas
Kerala	8
Uttarakhand	8
Jammu & Kashmir	7 (+1)
Himachal Pradesh	5
Meghalaya	4

(Parsing note: ~1,133 rows parsed cleanly of a reported ~1,150+; a handful of wrapped rows could not be state-resolved and are excluded from state counts. Raw CSV and script are in the evidence archive.)

The headline: the scam plazas were concentrated in exactly the states where plazas are densest — **UP (135 plazas in the registry) contributed the largest single block of the 42-plaza list (10)** — but the footprint is genuinely national: 14 states in the STF list alone, from Jammu to Assam.

13. Cross-walking the scam list against the registry

We matched the reported scam plazas against the official registry. Three results matter:

1. The scam plazas are ordinary plazas. Atraila Shiv Gulam appears in the registry as *Atralia SivGulam*, NETC code **320132**, district Mirzapur, on the “Varanasi to Hanumana pkg-III” section — no flag, no annotation, indistinguishable from the 1,132 honest plazas³⁷. The same for Patni Pratappur (NETC 536051, Shamli district) from the scam list. **There is no public marker in the government’s own data that anything was ever wrong at these plazas.**

2. The registry still lists them. As of the August 2025 edition — after the arrests, after the debarments — the plazas remain listed. We take no position on whether they should have been removed; but a citizen or journalist using only official data would never learn that a scandal had touched these locations.

3. The crosswalk is incomplete — by design, perhaps. The Baghpat scam plaza cannot be uniquely matched to one of the five Baghpat-district plazas in the registry from press names alone. Names in press lists (“Bagpat”, “Chikli”, “Shauly”) do not cleanly match registry names (“Jiwana”, “Baleni”, “Badagaom”...). **Matchable only via the concession contract or NETC code — neither public.** An RTI for the 42-plaza list with NETC codes is the obvious next step (see §17).

³⁷IHMCL, “National Highway (NH) Fee Plazas” list, August 2025 (PDF in evidence archive; <https://ihmcl.co.in/>).

14. The companion scams nobody counted

The registry also lets you see what *didn't* get counted. In 2025–26, GST-seat reporting and Hindi press documented a parallel SOP at UP plazas including Itaunja and Nawabganj (Lucknow): trucks running on **~900 fake vehicle-registration numbers per month** to dodge toll entirely — a different exploit of the same manual lane³⁸. Consumer complaints continued too: in March 2026, a Baghpat commuter produced a receipt **timestamped 8:46 PM for an 11:30 AM crossing** — the shadow-receipt signature persisting in the wild after the “fixes”³⁹. That plaza, ironically, is on the debarred-operator’s list.

The lesson for DPI watchers: every digital-public-infrastructure system that delegates the *edge* — the booth, the merchant, the CSC, the banking correspondent — to a private contractor inherits this exact risk: the edge can run unauthorised software, and the centre’s audits will read only what the edge chooses to write.

³⁸<https://upstox.com/news/business-news/latest-updates/how-govt-plans-to-plug-highway-toll-collection-loopholes-after-up-scam-exposure/article-154498/>

³⁹<https://www.tv9hindi.com/state/uttar-pradesh/cctv-camera-found-in-women-restroom-baghpat-toll-plaza-misdeeds-including-allegations-of-illegal-extortion-3708444.html>

Part V — The fix, and whether it fixes

15. What changed after January 2025

Control	When	What it does	Verdict
AI audit cameras + ANPR pilots	announced Mar 2025 ⁴⁰	Independent vehicle counting at lanes	The <i>right</i> control (independent count); deployment status unknown
Toll Remittance Alert System (TRAS)	tendered Apr 2023 ⁴¹	Alerts when agency remittances lag collections	Blind to this fraud class by design — it monitors <i>remitted</i> money; skimmed cash was never booked
Cash ban: FASTag/UPI-only lanes	10–11 Apr 2026	Ends the cash vector entirely	Kills the vector prospectively; transfers friction to consumers without tags
Unified ATMS software (RFP May 2026)	2026	One national plaza-software stack; spec contains no manual-lane concept at all ⁴²	Digitises the exception category itself; removes operator discretion
MLFF (barrier-free tolling) acquirer-bank model	pilot RFP Jan 2025 ⁴³	No booth, no operator, no cash — ANPR-based charging	Structural fix; per-plaza procurement model will scale slowly
STQC certification scheme for TMS (tamper-evident databases, audit trails)	scheme live ⁴⁴	Certifies the lane software itself	The control that would have <i>detected</i> the shadow app; unclear it was ever procured for plaza TMS

⁴⁰<https://www.gstseat.com/news/parliament-standing-committee-on-transparency-in-toll-collection-detailed-answer->

16. The procurement critique

Three tenders, read side by side, reveal the pattern:

- **TRAS (2023)** monitors remittances only — its data feeds are the agency’s own declarations. It is a watchdog fed by the watched.
- **Unified ATMS (2026)** is a 239-page spec with state-of-the-art cloud architecture — and **zero occurrences** of “manual lane”, “cash lane” or “handheld device”. The exception category has been designed away rather than instrumented. Good for fraud, but the residual risk migrates to ANPR mis-reads (fake and duplicate plates — already a documented racket⁴⁵) and to digital-exception fraud.
- **MLFF (2025)** is being procured **one plaza at a time** (“selection of acquirer bank... at Nemili”) — re-procurement as a scaling strategy is how you get a patchwork of vendors and a decade of transition.

And the deepest gap: **no signed, hash-chained, lane-event data integrity requirement** appears in any plaza-software procurement we reviewed. The TMS remains software the contractor buys from whomever it likes, certified (if at all) against no public register. STQC’s certification scheme — tamper-proofing, audit trails — exists on paper; we could not find a public record of a single plaza TMS certified under it. (*RTI filed to confirm.*)

17. What we still don’t know (and are filing for)

Our RTI applications (drafted, ready to file at rtionline.gov.in⁴⁶) target the remaining unknowns:

1. **FIR 0017/2025 with seizure lists** (SP, Mirzapur) — the primary evidence of what the software actually did.
2. **The 14-agency debarment file**: plaza-wise map with NETC codes, PBG ledger, penalty orders (NHAI) — the accountability ledger the HC gutted.
3. **TRAS status** (IHMCL): vendor, go-live date, alerts generated to date.

given-by-minister-gadkari-in-lok-sabha/

⁴¹IHMCL, “Selection of Agency for Design, Development, Implementation... of Toll Remittance Alert System (TRAS)”, RFP, 3 April 2023 (PDF in evidence archive).

⁴²IHMCL/NHAI, “Unified NHAI ATMS Software” RFP Vol-2, 26 May 2026, 239 pp (PDF in evidence archive).

⁴³IHMCL, “Selection of Acquirer Bank for FASTag-ANPR based Multi Lane Free Flow (MLFF) User Fee Collection at Nemili”, RFP, January 2025 (PDF in evidence archive).

⁴⁴<https://www.stqc.gov.in/en/toll-management-system-products-certification-scheme>

⁴⁵<https://upstox.com/news/business-news/latest-updates/how-govt-plans-to-plug-highway-toll-collection-loopholes-after-up-scam-exposure/article-154498/>

⁴⁶Workspace archive: Documents/research/FASTagScam/rti/RTI-DRAFTS.md.

4. **The Atraila concession chain** (NHAI RO Varanasi): who “Veltek” legally is, who the concessionaire was.
5. **STQC certification register** for plaza TMS products.

Part VI — What it means

18. For the consumer

If you travelled a non-FASTag vehicle through any of the 42 (now ~100) plazas between roughly 2023 and 2025, you probably **paid double toll in cash and received a receipt that never entered any government system**. Your legal remedy as an individual is effectively nil — the amount is small, the counterparty anonymous, and refunds for all faulty collections nationwide in 2024 totalled ₹12.55 lakh⁴⁷ — against a diversion measured in the hundreds of crores. The real loser is the exchequer; the real payer is you again, through taxes and future toll resets.

What you *can* do:

- **Keep FASTag active and linked** — the manual lane is the risk surface; digital lanes are settled and traceable.
- If you get a suspicious toll receipt (wrong timestamps, wrong plaza, amounts that don't match the schedule), report it to NHAI's helpline (1033) **and** to us — the Baghpat complaint shows the pattern persists.
- Support the boring fixes: independent lane counting, public concession data, certified lane software. That is where this stops.

19. For digital public infrastructure

This case is a template, and it should be studied beyond highways:

- **The edge is the attack surface.** UPI's QR codes, ONDC's seller apps, AgriStack's field agents — every DPI stack delegates its edge. An unauthorised app at the edge, showing the centre what the centre expects to see, is a *general* attack pattern.
- **Auditing the ledger is not auditing the world.** NHAI audited its TMS. The TMS was fed by the contractor's lane software. The shadow app never touched the TMS. **Reconciliation must anchor to physical events** (vehicle counts, not keyed records).
- **Punishment is not accountability.** NHAI's ₹100 crore crackdown evaporated in court because the evidentiary work hadn't been done. The ED probe is attempt two. In DPI governance, the sequence matters: forensics first, penalties second.

⁴⁷<https://www.gstseat.com/news/parliament-standing-committee-on-transparency-in-toll-collection-detailed-answer-given-by-minister-gadkari-in-lok-sabha/>

- **Deleting the vector beats policing it.** The April 2026 cash ban and the manual-lane-free ATMS spec do more than any audit ever did. The honest framing of India's cashless push is that **cash-outsided exceptions were the fraud surface all along** — but the honest cost accounting must also note that each deleted exception creates a new dispute class (plate spoofing, digital-notice fraud) that consumer bodies must now watch.

Appendix A — Methodology: the registry parse

- **Source:** IHMCL “National Highway (NH) Fee Plazas” list, PDF, August 2025 edition⁴⁸.
- **Extraction:** pdftotext –layout, then a Python parser over the row grammar S.No | NETC code | Name | State | District | Section | NH | PIU | RO, with next-line fallback for wrapped name cells and a special case for the Atraila row (“Atraila SivGulam”).
- **Result:** 1,133 rows parsed with state attribution; ~9 unresolved (wrapped rows). NETC codes verified by spot-check against known plazas (e.g., 320132 = Atraila SivGulam; 536051 = Patni Pratappur).
- **Crosswalk:** scam-plaza names from the Dainik Jagran 42-plaza list⁴⁹ were matched on normalised name + district. Unique matches: Atraila Shiv Gulam (320132), Patni Pratappur (536051), Madanpur (Assam). Baghpat and several others are ambiguous (multiple registry plazas per district); we do not force matches.
- **Limitations:** the registry names plazas by their fee-schedule names; press lists use local names. Concessionaire/agency attribution per plaza is **not** in the public registry — it requires RTI or concession documents.
- All scripts, the parsed CSV, and the source PDF are in the evidence archive (Appendix C).

Appendix B — The 42-plaza list (preserved)

From Dainik Jagran, 23 January 2025⁵⁰ — preserved here because the original article is already dead-linking:

Uttar Pradesh (10): Harro (Prayagraj) · Mungari (Prayagraj) · Umapur (Prayagraj) · Andi Lohra (Aza-mgarh, AK CC) · Bagpat (AK CC) · Faridpur (Bareilly) · Patni Pratappur (Shamli) · Atraila Shiv Gulam (Mirzapur) · Nainsar (Gorakhpur) · Chikli **Madhya Pradesh (6):** Jangabani · Mohtara (AK CC) · Shal-ibada · Shahdol · Gahra · Chikhlikala (Chhindwara) **Rajasthan (4):** Phulera (Jaipur) · Kadi Shehna (AK CC) · Shahpura · Shauly (operating company reported as “Anuvejan”, ≈ Innovision — debarred) **Gu-jarat (4):** Mokha (AK CC) · Rohisa (AK CC) · Okha Mandi · Kuchadi **Chhattisgarh (4):** Maharajpur · Mudiapara · Kumhari (Durg, AK CC) · Bhojpuri **Maharashtra (3):** Dashrakhed · Khani Bade · (third per

⁴⁸IHMCL, “National Highway (NH) Fee Plazas” list, August 2025 (PDF in evidence archive; <https://ihmcl.co.in/>).

⁴⁹<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

⁵⁰<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

list) **Jharkhand (3):** Nawasari · Turup (AK CC) · Tand Balidha **Punjab (2):** Dhulal · Jigha (AK CC) **Assam (2):** Madanpur (RK Jain Infra — debarred) · Balachera **West Bengal (2):** Gobari · Pashchim Madati **Jammu (1):** Van · **Odisha (1):** Kadali Garh · **Himachal (1):** Sanwara · **Telangana (1):** Jangaon

AK CC = AK Construction (debarred). Counts per Jagran; the outlet's state tallies differ slightly from other outlets (we preserve the primary list as published).

Appendix C — Evidence archive and negative results

This report is the analysis layer; the archive is the evidence layer. All 23+ press captures (with canonical URLs), five government PDFs (TRAS RFP, Unified ATMS RFP, MLFF RFP, plaza registry, ATMS training deck), the parsed registry CSV, RTI drafts and the master dossier live in the workspace under Documents/research/FASTagScam/.

What we looked for and did not find (as of 4 September 2026) — published for scientific honesty:

- No app-store, GitHub, code-hosting or certificate-transparency trace of ‘Mobdata’/‘Any’ or plausible variants.
- No public copy of the ED press release (site’s Umbraco API confirms 1,995 releases through 2 September 2026; none on this case yet).
- No public copy of FIR 0017/2025 or the charge sheet (RTI route).
- No public record of any plaza TMS certified under STQC’s scheme.
- The Lok Sabha answer PDF (USQ 1017) was unreachable from our network; the ministry’s reply is quoted via secondary reporting⁵¹.
- Caution: “MobData” (MobTech, Chinese analytics SDK) is an unrelated namesake. Several LinkedIn profiles named “Alok Kumar Singh” are unrelated persons; do not chase.

⁵¹<https://www.gstseat.com/news/parliament-standing-committee-on-transparency-in-toll-collection-detailed-answer-given-by-minister-gadkari-in-lok-sabha/>

Appendix D — Sources

ED/PMLA (Sep 2026): The Tribune⁵² · Daily Excelsior⁵³ · Times of India⁵⁴ · Mathrubhumi⁵⁵ · Bizz Buzz⁵⁶
STF case (Jan–Feb 2025): Times of India⁵⁷ · Aaj Tak⁵⁸ · Dainik Bhaskar⁵⁹ · Dainik Jagran^{60,61} · Navbharat Times⁶² · Amar Ujala⁶³ · OpIndia⁶⁴ · Hindustan Times (Gorakhpur)⁶⁵ **Official response (Mar 2025):** Lok Sabha answer via GST Seat⁶⁶ · Rediff/PTI⁶⁷ · UNI⁶⁸ · Daily Jagran⁶⁹ · Upstox/PTI analysis⁷⁰
Legal: India Legal Live (Delhi HC, 6 Jul 2026)⁷¹ **Policy & tech documents:** IHMCL TRAS RFP⁷² · Unified ATMS RFP⁷³ · MLFF Nemili RFP⁷⁴ · IHMCL plaza registry⁷⁵ · STQC TMS certification scheme⁷⁶ **Consumer**

⁵²<https://www.tribuneindia.com/news/india/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-not-bearing-fastag-stickers-ed/>

⁵³<https://www.dailyexcelsior.com/about-100-highway-plazas-including-some-in-jk-used-illegal-apps/>

⁵⁴<https://timesofindia.indiatimes.com/city/lucknow/ed-conducts-searches-at-14-premises-in-6-states-seizes-rs-1-03-crore-cash-in-toll-collection-fraud-case/articleshow/133732293.cms>

⁵⁵<https://english.mathrubhumi.com/news/crime/about-100-highway-plazas-used-illegal-apps-to-collect-toll-from-vehicles-without-fastag-ed-o0qf4hpy>

⁵⁶<https://www.bizzbuzz.news/crime/ed-raids-14-locations-across-india-in-toll-plaza-scam-seizes-over-rs-1-crore-cash-1403756>

⁵⁷<https://timesofindia.indiatimes.com/city/lucknow/mca-grad-behind-toll-fraud-arrested/articleshow/117498101.cms>

⁵⁸<https://www.aajtak.in/uttar-pradesh/story/toll-plaza-software-scam-fraud-worth-crores-stf-caught-3-accused-varanasi-uttar-pradesh-lclar-strc-2149511-2025-01-23>

⁵⁹<https://www.bhaskar.com/local/up/lucknow/news/toll-plaza-scam-3-gang-members-detained-including-software-engineer-133802942.html>

⁶⁰<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

⁶¹<https://www.jagran.com/uttar-pradesh/mirzapur-lucknow-stf-raids-toll-plaza-in-mirzapur-three-detained-on-charges-of-embezzlement-of-rs-120-crore-23871523.html>

⁶²<https://navbharattimes.indiatimes.com/metro/lucknow/crime/toll-tax-fraud-by-fake-receipts-through-software-at-42-city-of-india-news/articleshow/117503938.cms>

⁶³<https://www.amarujala.com/uttar-pradesh/mirzapur/notice-to-toll-plaza-managers-of-six-districts-in-120-crore-rupees-scam-in-mirzapur-2025-02-13>

⁶⁴<https://www.opindia.com/2025/01/nhai-orders-action-in-the-toll-plaza-scam-uncovered-by-the-up-stf-3-arrested/>

⁶⁵Hindustan Times, Gorakhpur edition, January 2025 (press capture in evidence archive).

⁶⁶<https://www.gstseat.com/news/parliament-standing-committee-on-transparency-in-toll-collection-detailed-answer-given-by-minister-gadkari-in-lok-sabha/>

⁶⁷<https://money.rediff.com/news/market/nhai-debars-14-toll-agencies-for-irregularities/23868620250321>

⁶⁸<http://www.uniindia.com/nhai-bans-14-agencies-for-irregular-activities-in-fee-collection-at-toll-plazas/india/news/3418421.html>

⁶⁹<https://www.thedailyjagran.com/india/toll-collection-fraud-nhai-bans-14-agencies-for-using-fake-software-to-collect-toll-illegally-rs-100-cr-security-confiscated-10225098>

⁷⁰<https://upstox.com/news/business-news/latest-updates/how-govt-plans-to-plug-highway-toll-collection-loopholes-after-up-scam-exposure/article-154498/>

⁷¹<https://indialegalive.com/constitutional-law-news/courts-news/delhi-high-court-sets-aside-nhai-action-against-toll-contractors-in-software-fraud-case/>

⁷²IHMCL, “Selection of Agency for Design, Development, Implementation... of Toll Remittance Alert System (TRAS)”, RFP, 3 April 2023 (PDF in evidence archive).

⁷³IHMCL/NHAI, “Unified NHAI ATMS Software” RFP Vol-2, 26 May 2026, 239 pp (PDF in evidence archive).

⁷⁴IHMCL, “Selection of Acquirer Bank for FASTag-ANPR based Multi Lane Free Flow (MLFF) User Fee Collection at Nemili”, RFP, January 2025 (PDF in evidence archive).

⁷⁵IHMCL, “National Highway (NH) Fee Plazas” list, August 2025 (PDF in evidence archive; <https://ihmcl.co.in/>).

⁷⁶<https://www.stqc.gov.in/en/toll-management-system-products-certification-scheme>

corroboration: TV9 Hindi (Baghpat, 5 Mar 2026)⁷⁷ **RTI drafts:** workspace archive⁷⁸

Report by CashlessConsumer research (agentic workflow, human-supervised). Feedback: cashlessconsumer@zo.computer. This report will be updated when the ED press release and RTI responses land. Companion page: cyber.cashlessconsumer.in.

⁷⁷<https://www.tv9hindi.com/state/uttar-pradesh/cctv-camera-found-in-women-restroom-baghpat-toll-plaza-misdeeds-including-allegations-of-illegal-extortion-3708444.html>

⁷⁸Workspace archive: Documents/research/FASTagScam/rTi/RTI-DRAFTS.md.